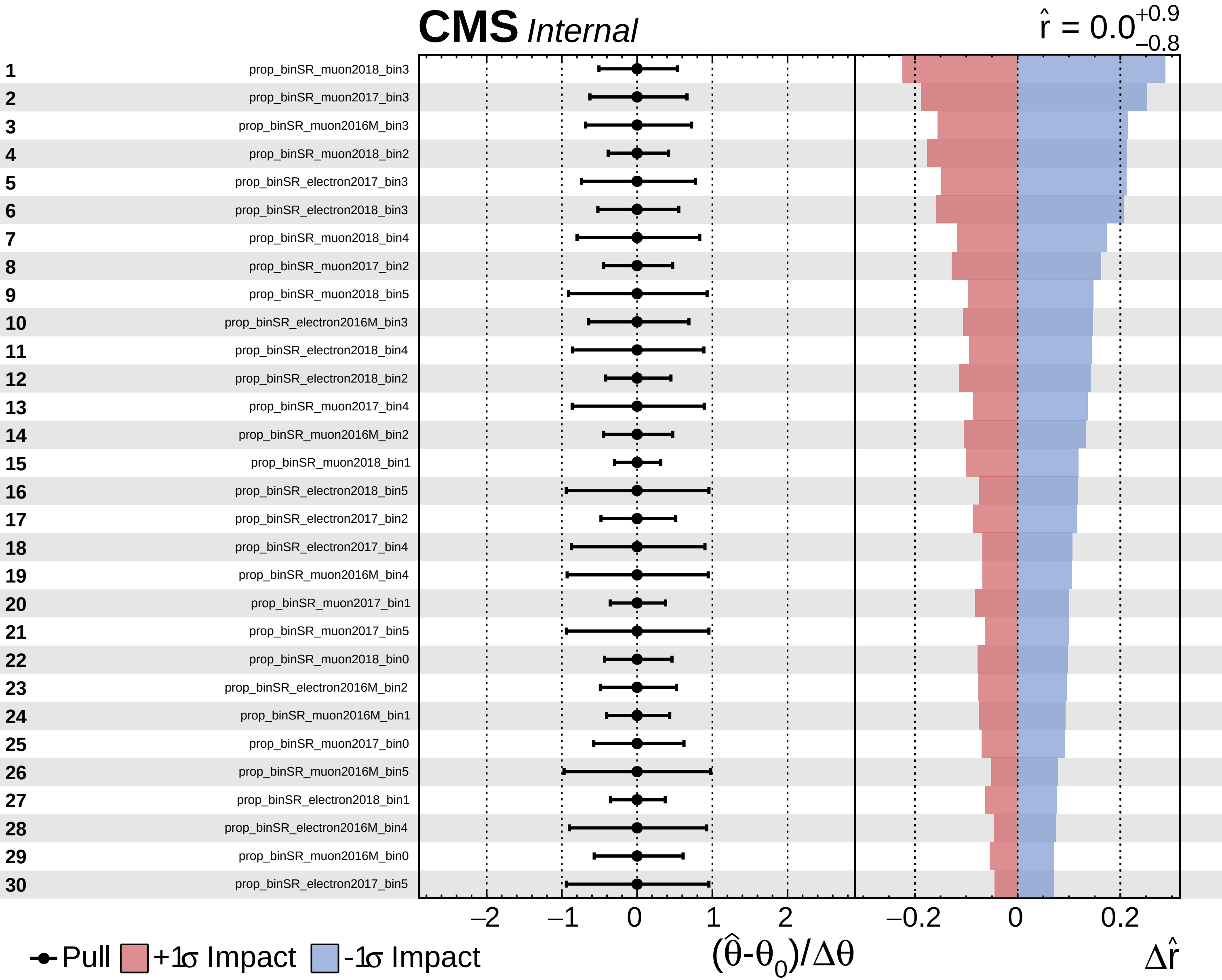
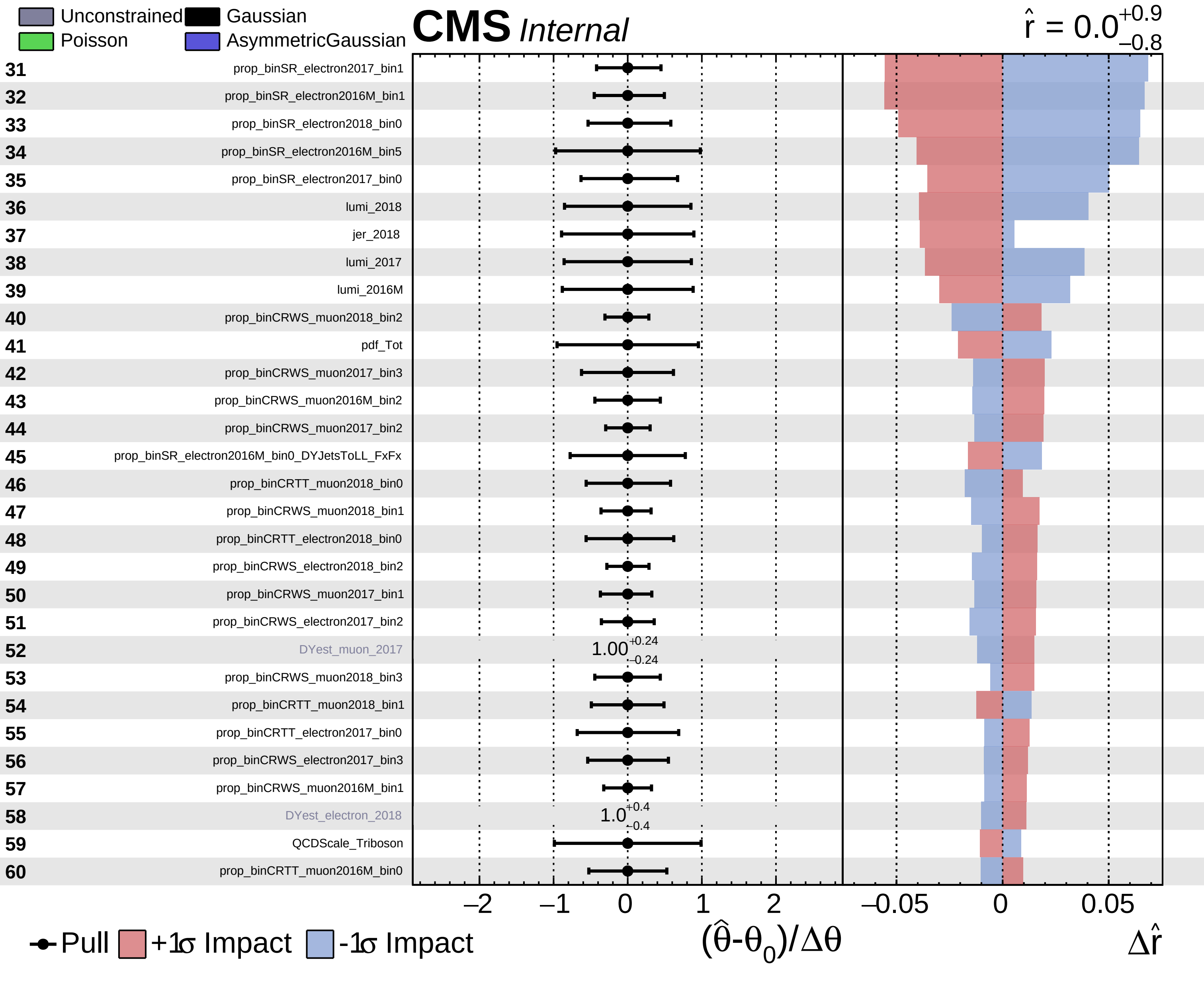
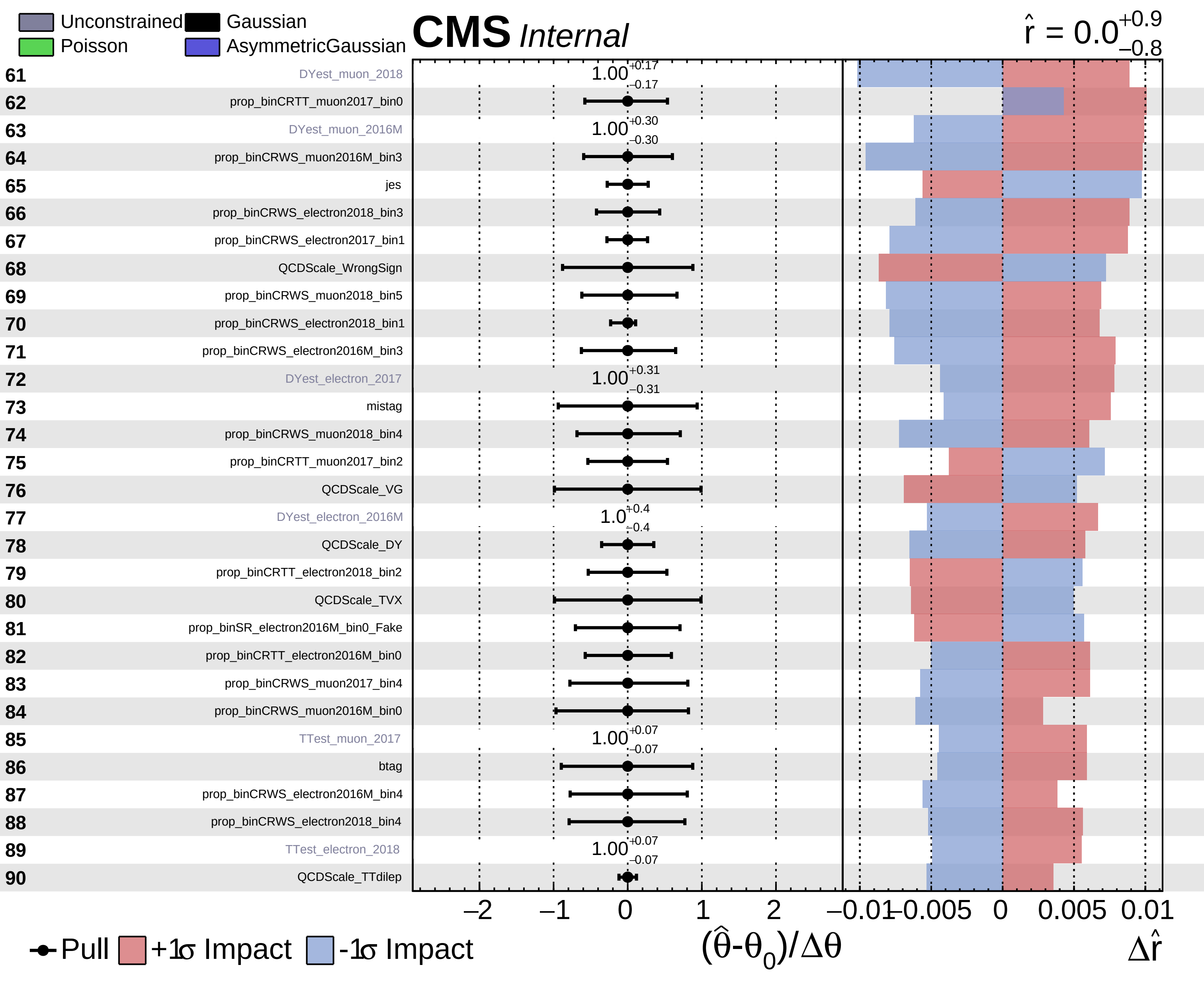


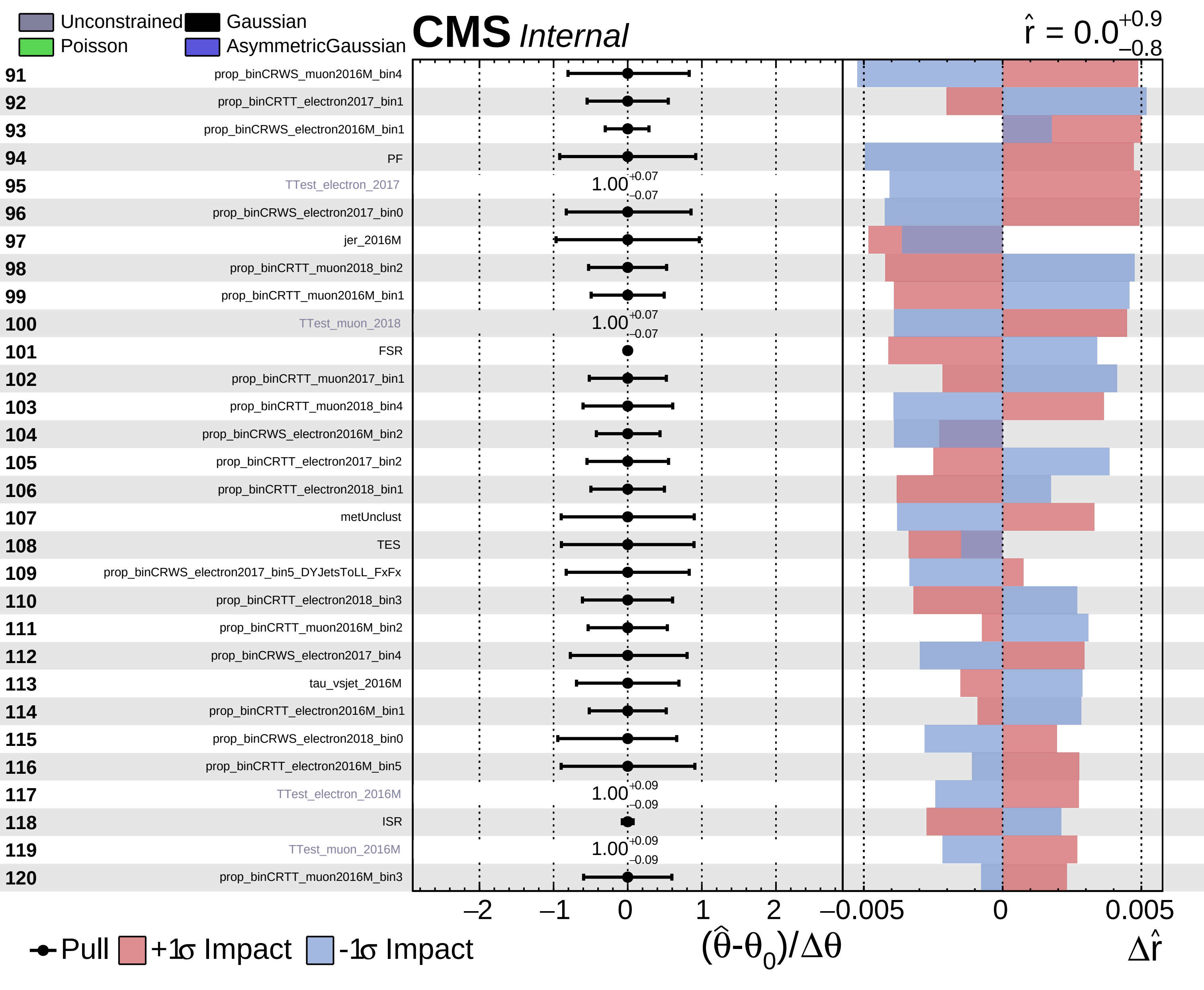
CMS Internal
$\hat{r} = 0.0^{+0.9}_{-0.8}$
1 prop_binSR_muon2018_bin3
2 prop_binSR_muon2017_bin3
3 prop_binSR_muon2016M_bin3
4 prop_binSR_muon2018_bin2
5 prop_binSR_electron2017_bin3
6 prop_binSR_electron2018_bin3
7 prop_binSR_muon2018_bin4
8 prop_binSR_muon2017_bin2
9 prop_binSR_muon2018_bin5
10 prop_binSR_electron2016M_bin3
11 prop_binSR_electron2018_bin4
12 prop_binSR_electron2018_bin2
13 prop_binSR_muon2017_bin4
14 prop_binSR_muon2016M_bin2
15 prop_binSR_muon2018_bin1
16 prop_binSR_electron2018_bin5
17 prop_binSR_electron2017_bin2
18 prop_binSR_electron2017_bin4
19 prop_binSR_muon2016M_bin4
20 prop_binSR_muon2017_bin1
21 prop_binSR_muon2017_bin5
22 prop_binSR_muon2018_bin0
23 prop_binSR_electron2016M_bin2
24 prop_binSR_muon2016M_bin1
25 prop_binSR_muon2017_bin0
26 prop_binSR_muon2016M_bin5
27 prop_binSR_electron2018_bin1
28 prop_binSR_electron2016M_bin4
29 prop_binSR_muon2016M_bin0
30 prop_binSR_electron2017_bin5
−2 −1 0 1 2
−0.2 0 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



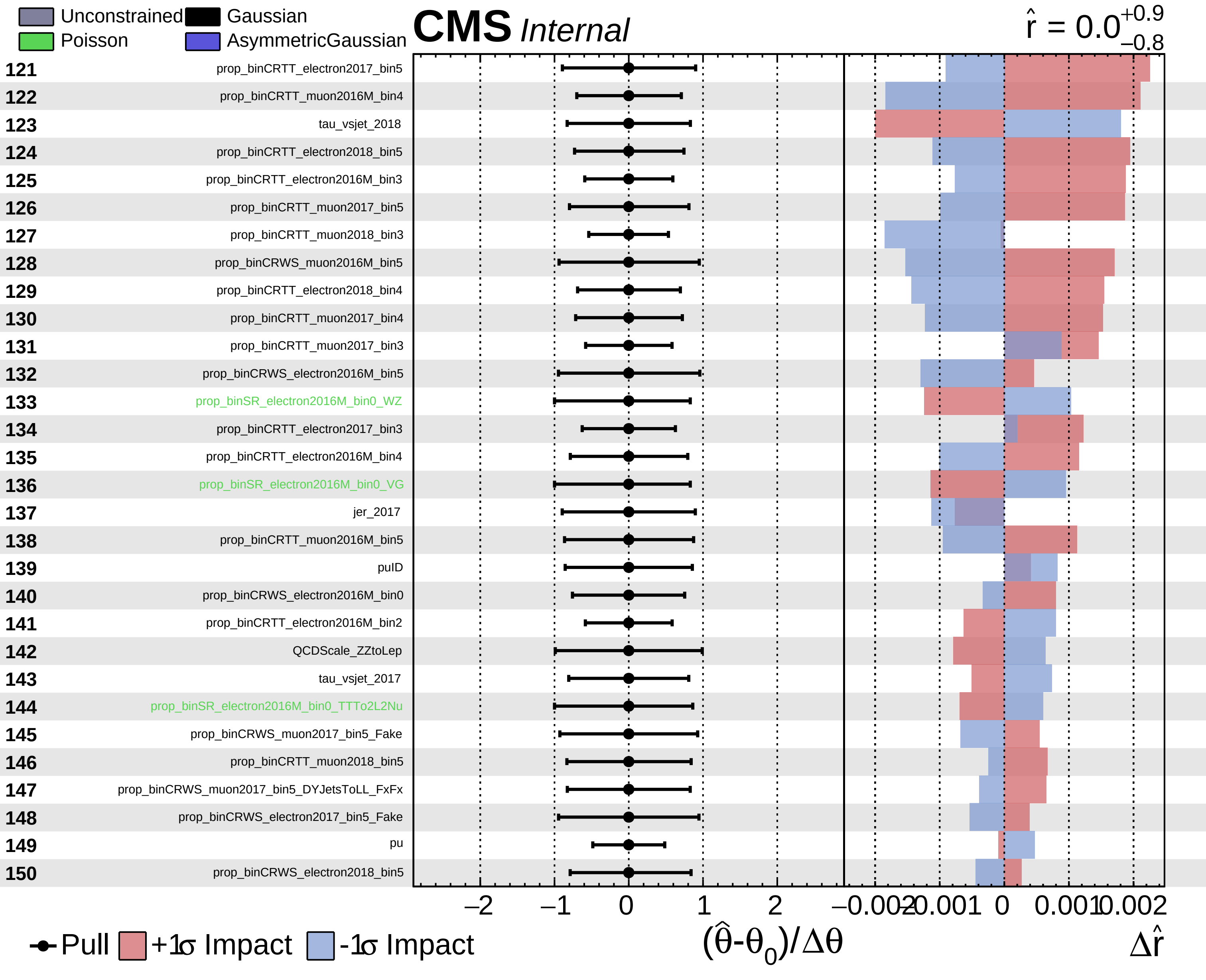
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.9}_{-0.8}$
31 prop_binSR_electron2017_bin1
32 prop_binSR_electron2016M_bin1
33 prop_binSR_electron2018_bin0
34 prop_binSR_electron2016M_bin5
35 prop_binSR_electron2017_bin0
36 lumi_2018
37 jer_2018
38 lumi_2017
39 lumi_2016M
40 prop_binCRWS_muon2018_bin2
41 pdf_Tot
42 prop_binCRWS_muon2017_bin3
43 prop_binCRWS_muon2016M_bin2
44 prop_binCRWS_muon2017_bin2
45 prop_binSR_electron2016M_bin0_DYJetsToLL_FxFx
46 prop_binCRTT_muon2018_bin0
47 prop_binCRWS_muon2018_bin1
48 prop_binCRTT_electron2018_bin0
49 prop_binCRWS_electron2018_bin2
50 prop_binCRWS_muon2017_bin1
51 prop_binCRWS_electron2017_bin2
52 DYest_muon_2017
$1.00^{+0.24}_{-0.24}$
53 prop_binCRWS_muon2018_bin3
54 prop_binCRTT_muon2018_bin1
55 prop_binCRTT_electron2017_bin0
56 prop_binCRWS_electron2017_bin3
57 prop_binCRWS_muon2016M_bin1
58 DYest_electron_2018
$1.0^{+0.4}_{-0.4}$
59 QCDScale_Triboson
60 prop_binCRTT_muon2016M_bin0
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



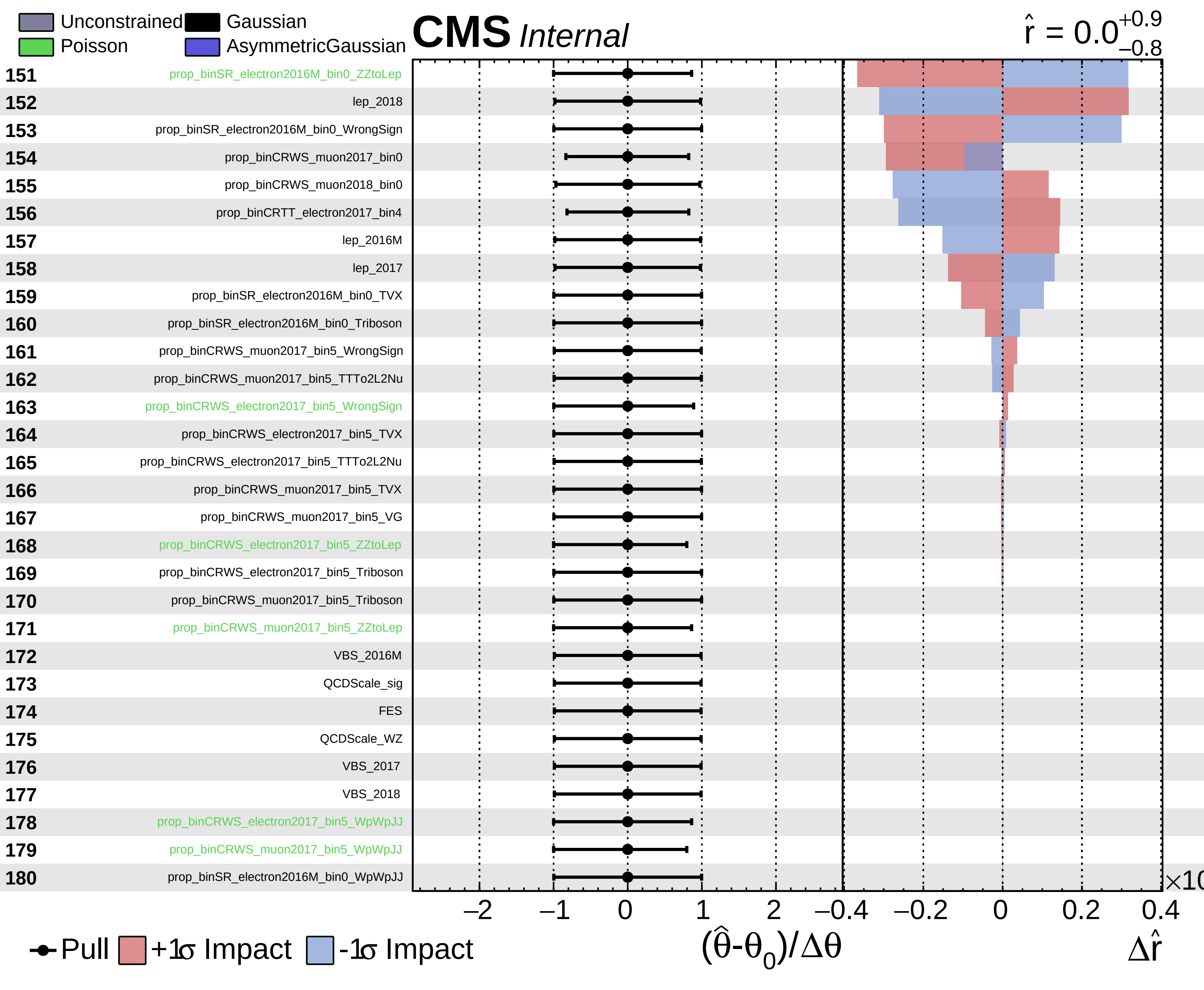
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.9}_{-0.8}$
61 DYest_muon_2018 $1.00^{+0.17}_{-0.17}$
62 prop_binCRTT_muon2017_bin0
63 DYest_muon_2016M $1.00^{+0.30}_{-0.30}$
64 prop_binCRWS_muon2016M_bin3
65 jes
66 prop_binCRWS_electron2018_bin3
67 prop_binCRWS_electron2017_bin1
68 QCDScale_WrongSign
69 prop_binCRWS_muon2018_bin5
70 prop_binCRWS_electron2018_bin1
71 prop_binCRWS_electron2016M_bin3
72 DYest_electron_2017 $1.00^{+0.31}_{-0.31}$
73 mistag
74 prop_binCRWS_muon2018_bin4
75 prop_binCRTT_muon2017_bin2
76 QCDScale_VG
77 DYest_electron_2016M $1.0^{+0.4}_{-0.4}$
78 QCDScale_DY
79 prop_binCRTT_electron2018_bin2
80 QCDScale_TVX
81 prop_binSR_electron2016M_bin0_Fake
82 prop_binCRTT_electron2016M_bin0
83 prop_binCRWS_muon2017_bin4
84 prop_binCRWS_muon2016M_bin0
85 TTest_muon_2017 $1.00^{+0.07}_{-0.07}$
86 btag
87 prop_binCRWS_electron2016M_bin4
88 prop_binCRWS_electron2018_bin4
89 TTest_electron_2018 $1.00^{+0.07}_{-0.07}$
90 QCDScale_TTdilep
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



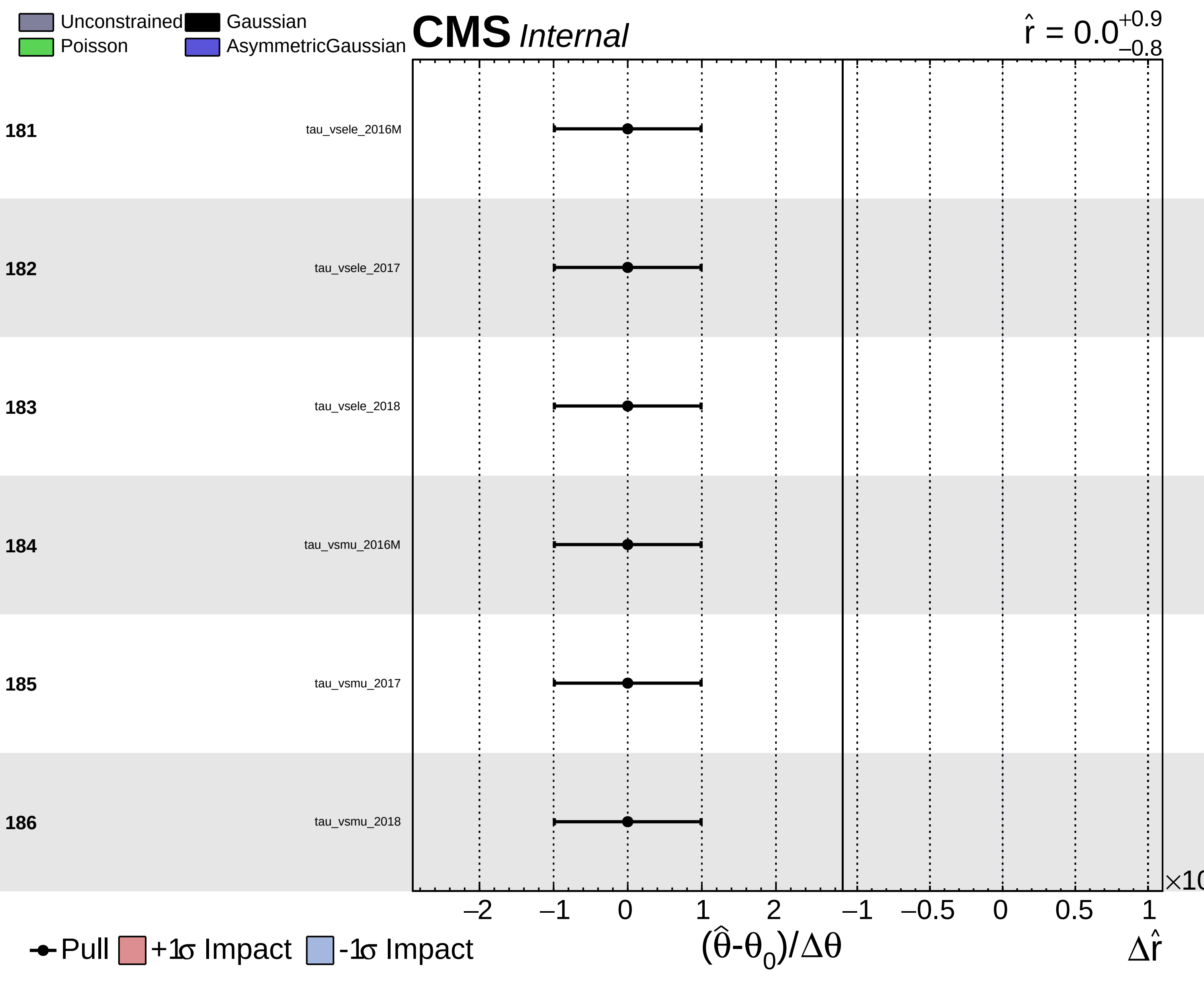
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.9}_{-0.8}$
91 prop_binCRWS_muon2016M_bin4
92 prop_binCRTT_electron2017_bin1
93 prop_binCRWS_electron2016M_bin1
94 PF
95 TTest_electron_2017 $1.00^{+0.07}_{-0.07}$
96 prop_binCRWS_electron2017_bin0
97 jer_2016M
98 prop_binCRTT_muon2018_bin2
99 prop_binCRTT_muon2016M_bin1
100 TTest_muon_2018 $1.00^{+0.07}_{-0.07}$
101 FSR
102 prop_binCRTT_muon2017_bin1
103 prop_binCRTT_muon2018_bin4
104 prop_binCRWS_electron2016M_bin2
105 prop_binCRTT_electron2017_bin2
106 prop_binCRTT_electron2018_bin1
107 metUnclust
108 TES
109 prop_binCRWS_electron2017_bin5_DYJetsToLL_FxFx
110 prop_binCRTT_electron2018_bin3
111 prop_binCRTT_muon2016M_bin2
112 prop_binCRWS_electron2017_bin4
113 tau_vsjet_2016M
114 prop_binCRTT_electron2016M_bin1
115 prop_binCRWS_electron2018_bin0
116 prop_binCRTT_electron2016M_bin5
117 TTest_electron_2016M $1.00^{+0.09}_{-0.09}$
118 ISR
119 TTest_muon_2016M $1.00^{+0.09}_{-0.09}$
120 prop_binCRTT_muon2016M_bin3
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.9}_{-0.8}$
121 prop_binCRTT_electron2017_bin5
122 prop_binCRTT_muon2016M_bin4
123 tau_vsjet_2018
124 prop_binCRTT_electron2018_bin5
125 prop_binCRTT_electron2016M_bin3
126 prop_binCRTT_muon2017_bin5
127 prop_binCRTT_muon2018_bin3
128 prop_binCRWS_muon2016M_bin5
129 prop_binCRTT_electron2018_bin4
130 prop_binCRTT_muon2017_bin4
131 prop_binCRTT_muon2017_bin3
132 prop_binCRWS_electron2016M_bin5
133 prop_binSR_electron2016M_bin0_WZ
134 prop_binCRTT_electron2017_bin3
135 prop_binCRTT_electron2016M_bin4
136 prop_binSR_electron2016M_bin0_VG
137 jer_2017
138 prop_binCRTT_muon2016M_bin5
139 puID
140 prop_binCRWS_electron2016M_bin0
141 prop_binCRTT_electron2016M_bin2
142 QCDScale_ZZtoLep
143 tau_vsjet_2017
144 prop_binSR_electron2016M_bin0_TTTo2L2Nu
145 prop_binCRWS_muon2017_bin5_Fake
146 prop_binCRTT_muon2018_bin5
147 prop_binCRWS_muon2017_bin5_DYJetsToLL_FxFx
148 prop_binCRWS_electron2017_bin5_Fake
149 pu
150 prop_binCRWS_electron2018_bin5
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.0 +0.9 −0.8
151 prop_binSR_electron2016M_bin0_ZZtoLep
152 lep_2018
153 prop_binSR_electron2016M_bin0_WrongSign
154 prop_binCRWS_muon2017_bin0
155 prop_binCRWS_muon2018_bin0
156 prop_binCRTT_electron2017_bin4
157 lep_2016M
158 lep_2017
159 prop_binSR_electron2016M_bin0_TVX
160 prop_binSR_electron2016M_bin0_Triboson
161 prop_binCRWS_muon2017_bin5_WrongSign
162 prop_binCRWS_muon2017_bin5_TTTo2L2Nu
163 prop_binCRWS_electron2017_bin5_WrongSign
164 prop_binCRWS_electron2017_bin5_TVX
165 prop_binCRWS_electron2017_bin5_TTTo2L2Nu
166 prop_binCRWS_muon2017_bin5_TVX
167 prop_binCRWS_muon2017_bin5_VG
168 prop_binCRWS_electron2017_bin5_ZZtoLep
169 prop_binCRWS_electron2017_bin5_Triboson
170 prop_binCRWS_muon2017_bin5_Triboson
171 prop_binCRWS_muon2017_bin5_ZZtoLep
172 VBS_2016M
173 QCDScale_sig
174 FES
175 QCDScale_WZ
176 VBS_2017
177 VBS_2018
178 prop_binCRWS_electron2017_bin5_WpWpJJ
179 prop_binCRWS_muon2017_bin5_WpWpJJ
180 prop_binSR_electron2016M_bin0_WpWpJJ
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.9}_{-0.8}$
181
tau_vsele_2016M
182
tau_vsele_2017
183
tau_vsele_2018
184
tau_vsmu_2016M
185
tau_vsmu_2017
186
tau_vsmu_2018
−2
−1
0
1
2
−1
−0.5
0
0.5
1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$